João André **Roque Costa**

MSc Graduate, Computer Science and Engineering

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Computer Science and Engineering MSc graduate focused on AI engineering, with hands-on experience building an **LLM** data-migration prototype using Claude SDK and Azure AI Foundry.
Combines **Python** automation, **Azure** and **Databricks** delivery, and academic depth in machine learning, autonomous agents, and scalable algorithms. Databricks-certified in Generative AI Application Deployment and Monitoring.

Professional Experience

Deloitte — *Coimbra, Portugal*
Tech Analyst — *Apr. 2026 - Oct. 2026*

- **Built** a cross-functional **LLM** prototype for data migration and synthetic data generation using Claude SDK, Azure AI Foundry, Azure Blob Storage, and MongoDB.
- **Delivered** an Azure subscription migration of a resource group and its Databricks and Azure Data Factory assets within the planned timeframe, coordinating Managed Instances and dependent services.
- **Prototyped** HR-report automation with Python and Playwright; contributed technical design to a client proposal for RAG- and OCR-based documentation automation.

Instituto Superior Técnico — *Lisbon, Portugal*
Teaching Assistant — *Oct. 2024 - Jul. 2025*

- **Led** and evaluated practical classes in Databases and Data Science, tracking student progress and giving constructive technical feedback.
- **Guided** Python data-analysis projects using pandas and matplotlib, adapting explanations to different audiences and levels of experience.

Education

Instituto Superior Técnico — *Lisbon, Portugal*
M.Sc. Computer Science and Engineering — *Sep 2023 - Nov 2025*

- Focused on **Artificial Intelligence** and data science, combining machine learning foundations with applied modeling and data-driven problem solving.
- **Relevant Coursework**: Deep Learning, Machine Learning, Autonomous Agents and Multi-Agent Systems, Optimization Algorithms, Cloud Computing and Virtualization.
- Thesis: **Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)**.
  - Developed a decomposition-based extension to Maximum Variance Unfolding for **nonlinear dimensionality reduction** on disconnected datasets.
  - Designed a parallelizable method that improved scalability for large datasets previously computationally infeasible to process.
  - Preserved local geometric structure while producing embeddings comparable to standard MVU.

Instituto Superior Técnico — *Lisbon, Portugal*
B.Sc. Computer Science and Engineering — *Sep 2020 - Jul 2023*

- Studied algorithms, distributed systems, operating systems, and database theory, building foundations for maintainable software and data-intensive systems.
- Applied **C/C++** in systems projects with emphasis on memory management, and designed relational schemas in **SQL**.

Extracurricular

Personal Software Projects (Self-Hosted)

- Built a self-hosted Linux system for **managing modular projects**, with per-machine resource management and a live monitoring dashboard.
- Developed a Discord **ELO matchmaking system** with PostgreSQL match history and Redis-backed caching and queue management.

Skills & Competencies

AI & ML	**LLM** application prototyping, RAG/OCR proposal work, machine learning, deep learning
Programming	**Python** (Claude SDK, pandas, NumPy, scikit-learn, TensorFlow, Keras, matplotlib, Playwright), C/C++
Data & Databases	**SQL**, PostgreSQL, MongoDB, MySQL, Redis, Microsoft SQL Server (SSMS), pgAdmin
Cloud & Data	**Azure**, Azure AI Foundry, Databricks, Azure Data Factory, Azure Blob Storage, Azure Managed Instances
Software Systems	Docker, Linux, Bash scripting, Git, REST APIs, Python automation with Playwright
Languages	Portuguese (Native), English (Fluent)
Collaboration	Cross-functional stakeholder communication, mentoring, technical presentation
Additional	LaTeX, ArchiMate